

University Saleh Boubnider Constantine 3.
Laboratory of General Anatomy.
Course intended for first-year medical students.

Faculty of Medicine. Department of Medicine.
Academic year 2025-2026.

GENERAL NEUROLOGY

PLAN

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II. GENERAL ORGANIZATION OF THE NERVOUS SYSTEM

A. THE CEREBROSPINAL NERVOUS SYSTEM

B. THE VEGETATIVE (AUTONOMIC) NERVOUS SYSTEM

I. INTRODUCTION:

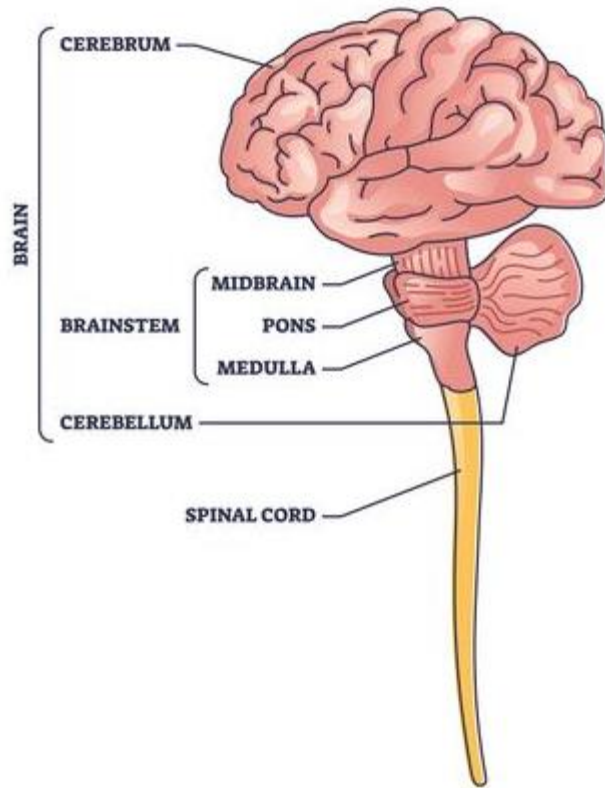
Neurology, or neuroanatomy, is the study of the nervous system.

The nervous system is defined as the set of structures that ensure the reception, integration, transformation, and transmission of information coming from both the environment and the organism itself.

II. GENERAL ORGANIZATION OF THE NERVOUS SYSTEM:

From an anatomo-functional point of view, the nervous system can be subdivided into two parts:

- The cerebrospinal nervous system, or the nervous system of animal life, which allows the life of relation — that is, all structures that permit voluntary interaction with the external world.
- The vegetative or autonomic nervous system, which governs vegetative life. It includes nervous structures in the brain, vegetative columns in the spinal cord, sympathetic ganglia, and vegetative nerves; it is essential for maintaining vital functions (cardiac and respiratory rhythms, thermal control, deep visceral pain, etc.).



Anatomie of nervous system

A. THE CEREBROSPINAL NERVOUS SYSTEM:

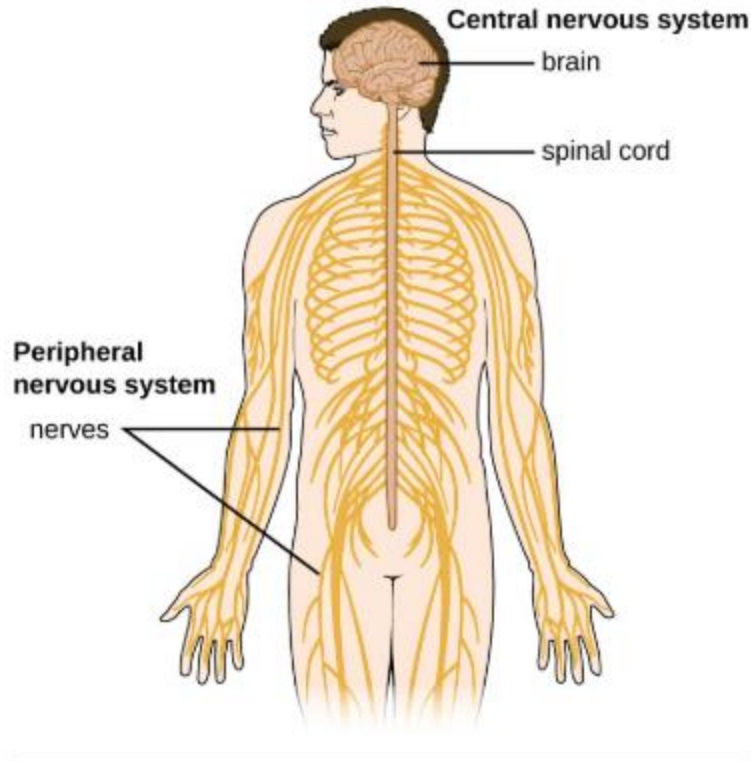
It includes:

1. THE CENTRAL NERVOUS SYSTEM (CNS) OR NEVRAXIS OR CEREBROSPINAL AXIS:

It integrates numerous stimuli which it analyzes, interprets, and memorizes. It supports major cognitive functions. It contains all centers of relational life (the conscious system acting on the environment) as well as centers of the autonomic system (visceral innervation and regulation of homeostasis).

The CNS includes:

- The Encephalon (brain + brainstem + cerebellum)
- The Spinal Cord



Cerebrospinal nervous system

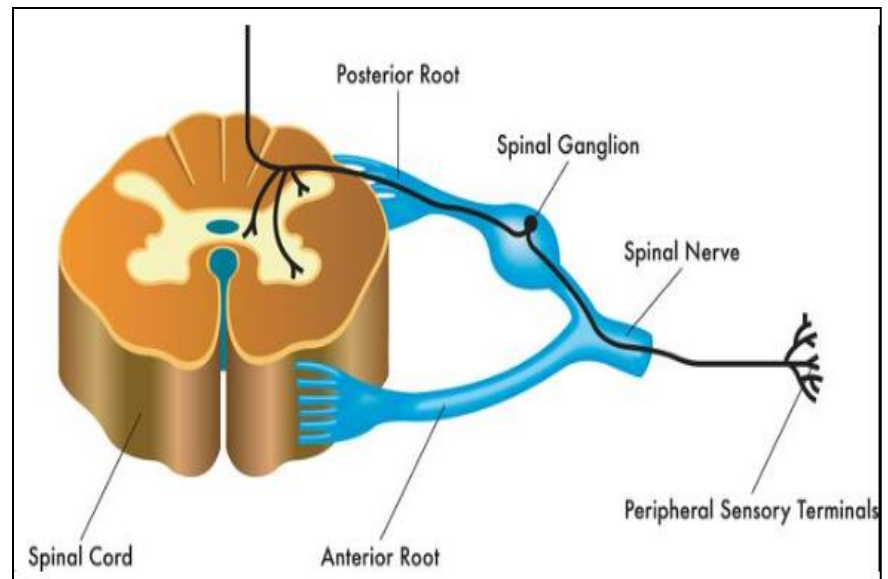
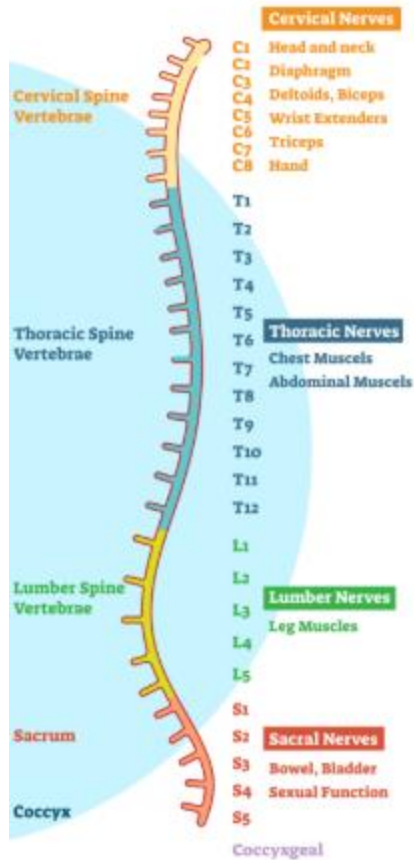
2. THE PERIPHERAL NERVOUS SYSTEM (PNS):

It connects the CNS to receptors and effectors. It consists of nerves arising from the neuraxis. Nerves are white cords connecting different parts of the body to the central nervous system.

- Nerves connected with the spinal cord are called spinal nerves.
- Those continuous with the brain are called cranial nerves.
- Both cranial and spinal nerves are arranged in symmetrical pairs on either side of the central axis.

Spinal nerves: there are 31 pairs, including 8 cervical nerves, 12 dorsal nerves, 5 lumbar nerves, 5 sacral nerves and 1 coccygeal nerve.

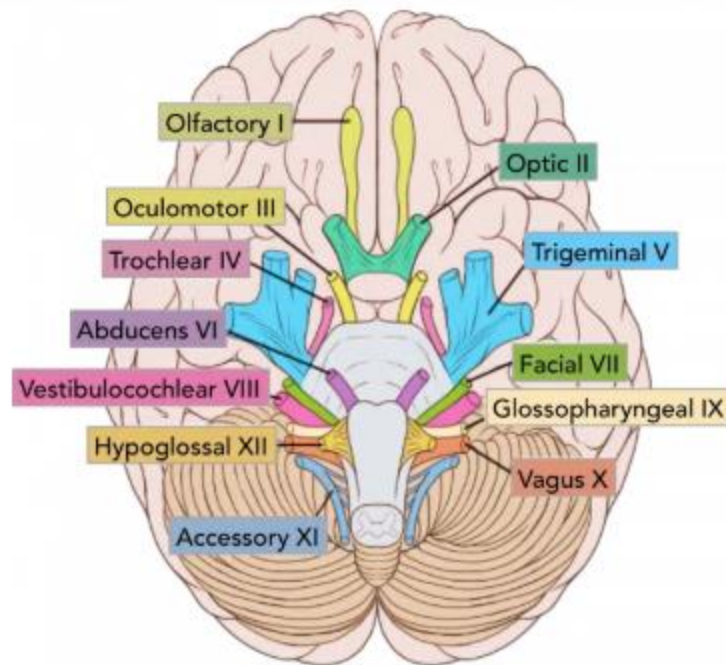
Each spinal nerve detaches from the spinal cord by two roots: one anterior motor nerve and the other posterior sensory nerve. Along the path of the sensory nerve root is a ganglionic enlargement: the spinal ganglion.



Spinal nerves

Cranial Nerves (12 pairs):

1. Olfactory nerve (I)
2. Optic nerve (II)
3. Oculomotor nerve (III)
4. Trochlear nerve (IV)
5. Trigeminal nerve (V)
6. Abducens nerve (VI)
7. Facial nerve (VII)
8. Auditory (Vestibulocochlear) nerve (VIII)
9. Glossopharyngeal nerve (IX)
10. Vagus nerve (X)
11. Spinal accessory nerve (XI)
12. Hypoglossal nerve (XII)



Cranial nerves

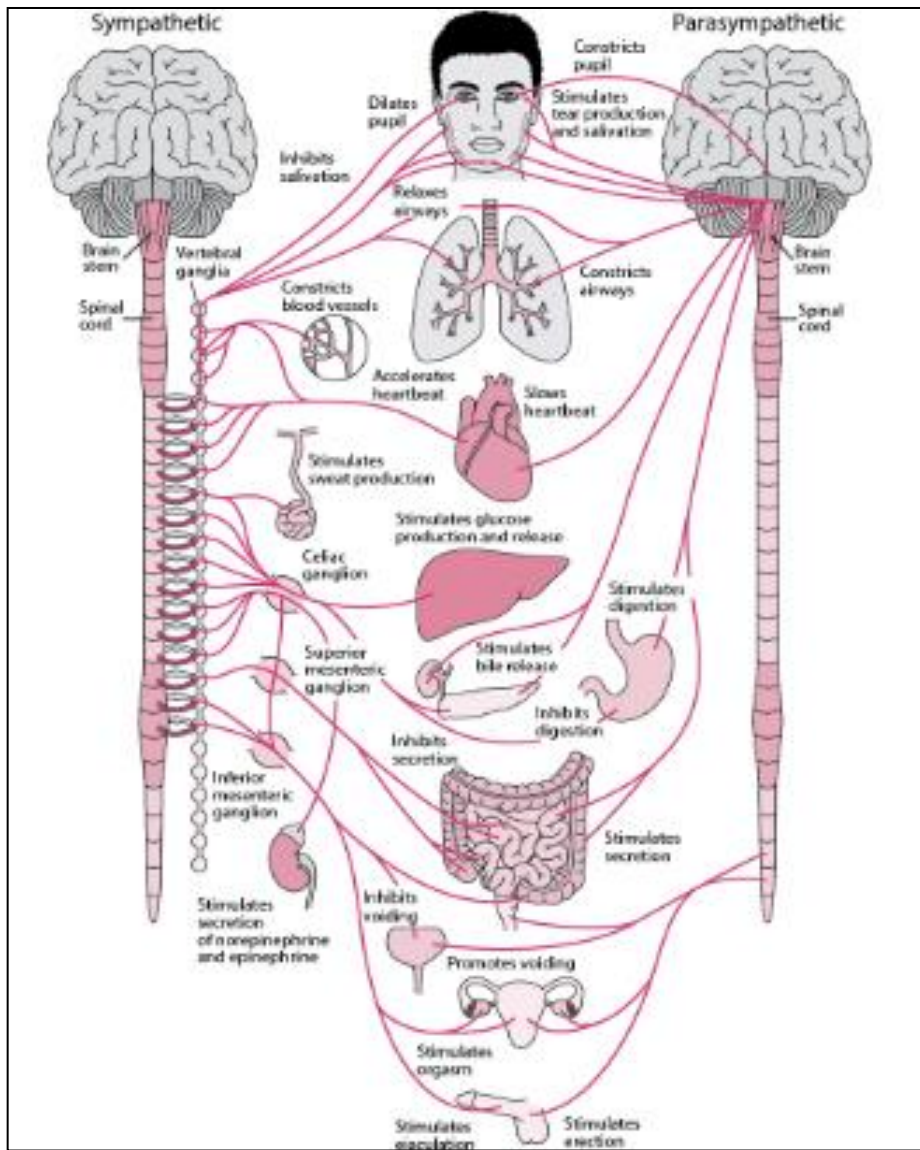
B. THE VEGETATIVE (AUTONOMIC) NERVOUS SYSTEM (ANS)

The autonomic (vegetative or visceral) nervous system controls the constancy of the internal environment (homeostasis) and innervates the viscera. It is composed of two antagonistic but complementary systems:

1. The sympathetic system: noradrenergic, predominates in stressful situations (fear, major bleeding, shame, etc.).
2. The parasympathetic system: cholinergic, predominates in basal conditions, ensuring the normal functioning of the body at rest.

Each includes a central part located within the cerebrospinal centers (neuraxis) and a peripheral part composed of nerves and ganglia.

The sympathetic nervous system includes a chain of ganglia arranged on each side of the vertebral column, connected by intermediate nerve cords, forming the sympathetic chain.



Vegetative nervous system

SENSE ORGANS

1. THE ORGAN OF VISION:

Consists of the main organ—the eye—and its nerve (optic nerve), as well as accessory organs such as ocular muscles, orbital fat, lacrimal apparatus, and eyelids.

2. THE VESTIBULOCOCHLEAR ORGAN:

Located in the petrous part of the temporal bone and composed of two distinct parts:

- The auditory organ: includes the external, middle, and anterior part of the inner ear, and the cochlear labyrinth.
- The equilibrium organ: represented by the posterior part of the inner ear and the vestibular labyrinth.

3. THE GUSTATORY ORGAN:

The mouth is the initial portion of the digestive tract, intended for mastication, taste perception, and salivation, as well as articulation of sounds. The tongue is the main organ of taste, consisting of a fixed root and a free body.

4. THE OLFACTORY ORGAN:

A chemoreceptor detecting odor particles, located in the postero-superior nasal cavity. The yellowish olfactory mucosa adheres to the superior nasal concha and the upper part of the nasal septum (about 1 cm).

5. THE ORGAN OF TOUCH:

Corresponds to all cutaneous sensory receptors distributed throughout the skin.

NOTE: LIMB INNERVATION

At the cervico-thoracic and lumbosacral levels of the spinal cord, the anterior (ventral) branches of the spinal nerves form anastomotic networks called plexuses, from which the peripheral nerves of the limbs originate.

Two main plexuses are distinguished:

- The brachial plexus (upper limb)
- The lumbosacral plexus (lower limb)

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